Final Design Report

The Printato

Prepared for:

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- Evanston, Illinois 60208

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Introduction

Project Description

The group that worked on this project is composed of four undergraduates from Northwestern University. In this project, the client, Dr. Alice Boone, requested that we make a unique hand printing press as part of our Design Thinking and Communication course (Appendix A). The hand printing presses of today are very effective, but that also introduces another problem. They are usually so efficient at doing their job that there is no room for error or creativity. The image that it is supposed to be produced is done so without any differences. While this might be the desired result when mass producing images, innovations in art can occur through serendipitous accidents, which is the main focus while producing these new hand printing presses (Appendix A). Additionally, it was requested that we create a press that is made out of a “daily object,” something that a person would be able to find around their house. This would demonstrate a solid connection between engineering and art. To that end, our design focused on convenience and adaptability so that there would be few limitations in what a user could create.

Printing Types

There are plenty of different ways to print and just as many materials one can print with. From a regular hand press using ink to printing different types of proteins (Appendix H), the handpress is a versatile machine that has a variety of applications. In this project, our team will aim to print with paint and ink, and our design will give the user a variety of different backgrounds to print with.

Report Overview

This report will go through all of the specifics behind our team’s hand printing press. It will introduce the people that use it and the client. Afterwards, there will be a discussion about the parts of the design and all of the reasons behind them. Finally, it will end with a discussion about the drawbacks of the design and how it can be improved with more time and resources.

Users and Requirements

In terms of users we expect many of the users to be children interested in arts and crafts. Since the project calls for a press that can be made at home, we assume that a lot of children looking for arts and crafts projects will be using our design. Since much of our audience will be children, there were many design considerations we had to make to ensure that our design suited our user.

The most important requirement, especially when dealing with children, is safety. Children can be careless in their actions and could accidentally hurt themselves with our printing press if we did not design it properly. In terms of the design, the need to maximize safety means that there cannot be sharp edges, places where fingers can be easily pinched or excessive pressure from the press itself. This limits what materials we can use on our design and what we could do with our design itself. For instance, we could not use thin sheet metal as the base for our printing block since sheet metal can have sharp edges that cut skin. Another compromise we had to make was with the extending arm of our press. We initially wanted to use a telescoping rod but decided against it since it could easily pinch a child's skin. Instead, we opted for a drawer rail that had less of a risk.

Another requirement when designing for children is simplicity. Since children have limited comprehension skills and attention span relative to teenagers and adults, the printing press must be simple to operate. This means that when we were brainstorming ideas, we did not consider any ideas we considered too complex since that would make our design inaccessible to a large audience, which is why we settled on a basic hinge design. When designing our rotating and extending printing arm, we settled on a very simple design with drawer rails and a lazy susan to bring versatility and simplicity to our hand press.

The final requirement is more catered to a general audience. Since some of our audience will also be adults with an interest in printing as an art form, we need a design that is versatile enough to create multiple designs. This may involve attaching different objects to the print block, using different surfaces as a print block or allowing for fast swapping between objects. In this area, we made several design considerations. We decided to use velcro on our printing block so users could swap out what was on the block if they wanted to create a new design. We also decided to utilize drawer railings and a lazy susan so that the printing arm could have a wider range of motion and print on multiple pieces of paper. All of these design considerations contribute to a better user experience, where users can print any design they desire.

Design Concept and Rationale

The Printato (Figure 1) is a solution tailored for anyone interested in print making, aiding the creation of prints with efficiency and at a fast pace while minimizing potential mess. Central to its design is a potato masher with a foldable handle, made to afford users optimal grip and the most comfortable position when print making. A piece of velcro is added onto the head of the potato masher and serves a dual purpose: fixing the desired design for printing securely and swiftly changing the designs. The potato masher is attached to a lever using a rail to increase the mobility of the device. The foldability of the lever which is mounted onto a lazy susan mechanism via a hinge, allows for the

actual print making. The lazy susan mechanism is attached to a base using a railing, providing stability and enabling full 360° rotational motion of the lever press as well as its forward and backward movement. Such versatility gives the user flexibility in selecting precise locations for printing, enriching the creative process.



Figure 1. The Printato Prototype

The following sections describe the above mentioned individual components of the device in more detail along with the rationale behind them.

Foldable Handle Potato Masher

Specifications

The potato masher is 1.6 x 3.9 x 11.9 in and weighs 2.6 oz. It is constructed of a plastic triangular head, metal handle and a rubber grip on the handle. The handle is attached to the head on a hinge, allowing foldability. A self-adhesive piece of velcro is securely glued to the potato masher head. The user sticks their design to the velcro and inks it, then operates the press gripping the handle.

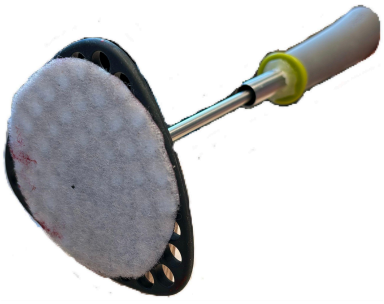


Figure 2. The potato masher

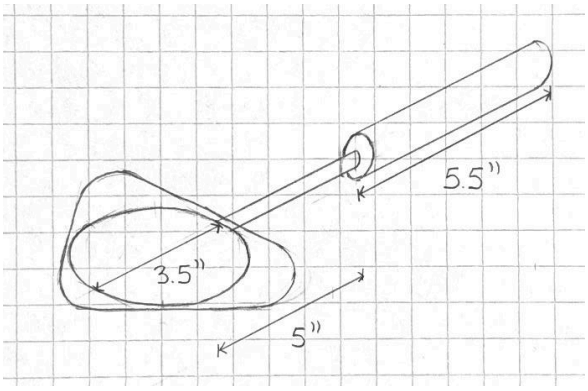


Figure 3. Dimensions of the potato masher

Rationale

The potato masher is used as a stamp for the design. The velcro on the head of the potato masher allows for the interchangeability of the designs specific to user's desires, as the user placed a strong emphasis on the variability of possible prints during the user interview (Appendix A). By gluing velcro to the designs the user wants to print, they can easily stick and remove them from the potato masher, allowing for the quick and easy changeability. The ability to remove the designs from the press itself also enables the user to easily clean them if wanted, which was another requirement made by the user during the interview (Appendix A). The choice of using a potato masher specifically stemmed from the user observation (Appendix B), where it was noticed that the user had to apply a lot of pressure using their hand, which was not only difficult for the user but also caused the non-uniform application of pressure. Using a potato masher solves both of these problems. Applying the pressure on the handle requires less force as that force is then redistributed through the head of the potato masher and due to said redistribution, the pressure is uniformly distributed.

The handle of the potato masher has a rubber grip, whose main purpose is to provide a comfortable grip of the hand press as it is the part that is held when operating the device. The rubber is soft and thick which minimizes the impact of the counter-pressure of pressing down on the hand. Such material also allows for a stronger grip, making it not slip from the users hand during operation.

The handle is also foldable, allowing for its forward and backward motion. This is beneficial to the overall simplicity and comfort of the press operation. The foldability enables the press to adjust to specific arm positions of the user. This allows the user to determine the most comfortable position for pressing down according to their preferences, as the uncomfortable operating positions were something that was noticed during the user observation (Appendix B).

Lever

Specifications

The lever is a 7.25 x 4.50 x 0.75 in wooden board with a 3.5 x 1.5 in metal hinge screwed to its bottom and a 1.75 in wide metal rail screwed to its back. The metal rail is extendable, extending from 9.0 to 15.0 in and allowing for the potato masher attached to its end to be moved back and forth, reaching all the parts of the paper on which it prints.



Figure 4. The lever

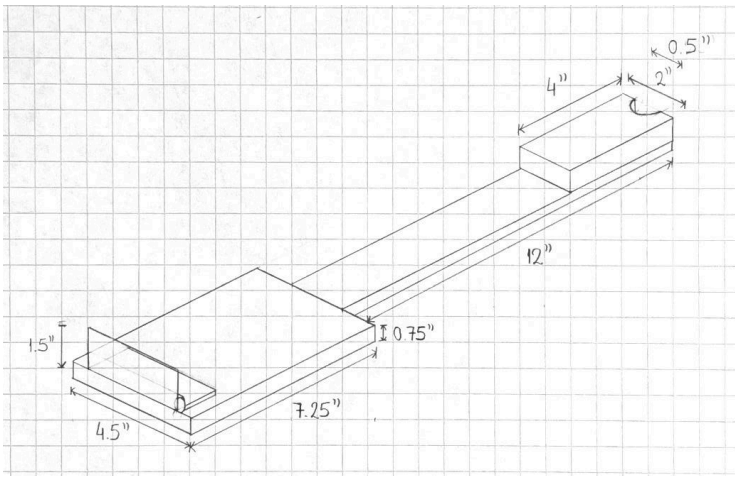


Figure 5. Dimensions of the lever

Rationale

The lever with the hinge allows for printing due to its foldability, while also keeping the potato masher stable and aimed at the printing surface. Something that was noticed

during the user observation (Appendix B) is that the pressure applied by hand is imprecise as the user has to be observant of the hand placement on the design that they are printing and this very hand placement often impacts the quality of the print itself. Using a lever to control where the pressure is applied, or where the design is printed, not only eliminates the issue which arises due to the imprecise hand placement but also makes it easier for the user to see where exactly they are printing.

The rail on the back of the lever expands the possibilities of print placements. As it can extend up to 6 inches, it gives the user more flexiblity when determining the exact place of their print on the paper or the surface they are printing on, enabling them to simply move the stamp, the potato masher, instead of moving the paper itself.

Base

Specifications

The base is made out of two components connected via a metal rail. The first component is the base board, a 8.0 x 7.5 x 0.75 in plate made of wood, which is supported by four 1.5 x 1.5 x 0.75 in posts. The top surface of the base board is covered with an adhesive plastic sheet to protect the wood from staining. The base board is connected to a 4.75 x 4.5 x 0.75 in wooden plate supported by two 1.5 x 1.5 x 0.75 in posts by an extendable metal rail. The distance between the two wooden plates is 4.25 in when the rail is not extended and reaches up to 5.5 in when extended. The second plate has a 360° rotating 4” plastic lazy susan mechanism screwed on to it which is used for attaching the lever to the base. The base is the foundation of the Potato Masher Lever Press, giving the device its stability and indicating the placement of the printing paper to the user.

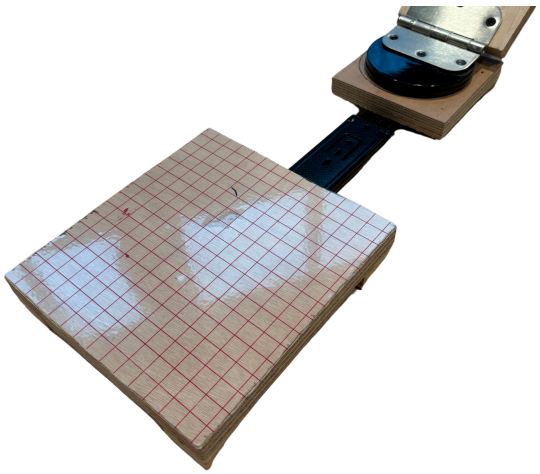


Figure 6. The base

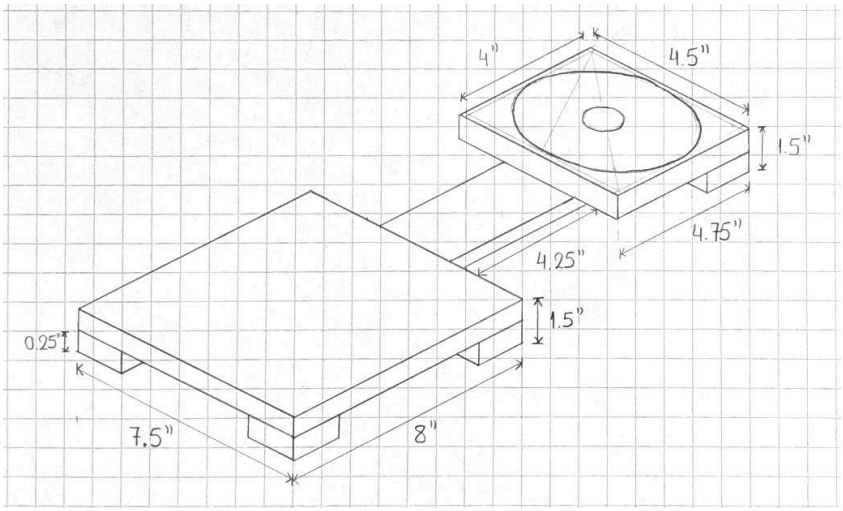


Figure 7. Dimensions of the base

Rationale

The base is the component makes the press stable and upright standing. It is the action point of the lever which enables its proper functioning.

The base board serves a purpose of holding the paper or any other desired printing material. If said material is smaller than the actual board, it will act as a protective surface against the paint splashing, allowing for a mess-free printing process, which was deemed important after the user observation (Appendix B). For the same reason, the base board is covered with an adhesive plastic wrap, allowing the user to simply wipe down the board if the paint gets transferred to it. The foil has a square grid on it, enabling the user to align the paper with the press if wanted.

The rail connecting the two plates allows the user to adjust the position of the lever and the potato masher compared to the base board. It also makes the device more compact and easier to store, which was a user requirement discussed during the user interview (Appendix A).

The lazy susan mechanism attached to the second plate is the connection between the lever and the base. It allows 360° rotation of the lever and the potato masher, enhancing the variability of the prints. The mockup testing (Appendix C) done on a mockup without this feature showed that the rotation of the lever would not only increase the amount of print possibilities but also enhance the simplicity of its use, as the user would be able to print on the desired part of the paper by moving the lever instead of the paper.

Attachments

Concept

The attachments are the part of the device that is creating the print. They are made of whatever the user wants to print with and attached to the potato masher via a velcro strip. As an example, multiple attachments were made using bubble wrap, cardboard and pom poms stuck to foam core. Additionally, multiple self adhesive velcro strips are provided with the device for the user to stick on their own designs.

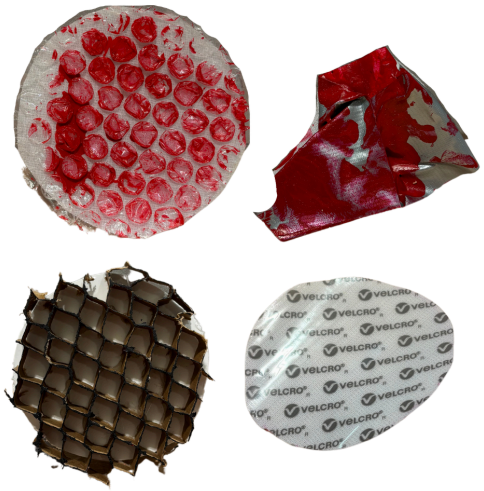


Figure 8. Attachments

Rationale

Giving the user the ability to change attachments and create the designs of their own is what this press is ultimately about. The user is provided with velcro strips that enable the attachment to be stuck to the press and easily remover. The very design of the attachments is entirely up to the user and can be any shape or size, as long as the velcro is able to stick to it. This enhances the creative process as it allows the user to experiment with various designs and print types.

Directions for Future Development

One limitation of this design is that the user cannot control how hard they are pressing the applicator onto the print. If the user wanted to reproduce a print multiple times with some sort of control and consistency, they cannot do this with the current design. One way to resolve this issue is by adding different levels of tension onto the handle. Possibly by using Nylon string or rubber bands, the individual using the press could tie off different amounts of this material to increase or decrease the tension of the handle. This tension would vary how much you could press the handle down and add a quantitative way to measure how much force one is using to press down the applicator.

Another limitation of this design is the attachments' inability to be used repetitively. In other words, the current attachments made have a "lifespan" and cannot be used forever. Once paint is attached to the material, it is difficult to clean the attachments depending on the material used. Future prints could be influenced as the dried paint on top of attachments could distort the textures and designs to a point that is eventually unusable. By using materials that can be cleaned, this problem can be resolved. Since this alternative is unlikely, however, as artists will want to use this device with various materials that cannot be easily cleaned, the best option is to continue creating new applications with different pieces of velcro rather than reusing old ones.

It is also important to note that the press is a bit difficult to construct for the average civilian, especially a child. Dr. Boone preferred that the press be made easily so that others can make it themselves, and this design does not quite meet that requirement as many components had to be screwed in and manipulated. To counteract this issue, it is possible that the main body of this device could be replaced with a panini press or even a book. Although the range of motion to print would be limited with this replacement, it would allow for the press to be more easily made.

In an overall sense, another iteration of our team's prototype could display profound improvements. Knowing this, it is worthy to acknowledge that some of these developments may stray from Dr. Boone's guidelines for how the project should be made. For instance, the idea of creating a cylindrical base board rather than a rectangular one would allow the artist to create multiple prints at once as the lazy susan enables the potato masher to rotate 360 degrees. This would be beneficial in making the printing process faster, but it would make the press created larger and would call for prints that are not only 4 by 6 inches. Developments like these advance the concept of variability and experimentation within the printing operation, but they tend to go beyond the range of the requirements Dr. Boone set out for our class.

Conclusion

The goal of our design is to make printing easy and reduce the reproducibility of prints. The design places importance on ease of use to allow children to be able to use this press without much outside help. The design utilizes:

- A lazy susan joint that can turn 360 degrees
- Drawer slide rails with a pull out length of 7.76 inches
- A potato masher to easily change the attached printing surface

These features make it simple to change the image printed on the paper while making the design easy and fun to use. Additionally, All of the sharp screws have been covered and the press is elevated to avoid a user's hand being slammed.

Appendix A

Project Partner Interview Summary

Introduction

In this research, our team engaged in an insightful interview with Dr. Alice Boone, our project partner and end user, as part of our primary research strategy. The focal point of this interview was to gain comprehensive insights into Dr. Boone's unique needs and expectations concerning the design of a hand-printing press crafted from common kitchen utensils. The main goal of this investigation was to aid the development of a custom hand-printing press that fully fits Dr. Boone's needs and utilization settings while also satisfying technical requirements. By delving into the specifics of her requirements and understanding the nuanced aspects of her creative workflow, this research seeks to inform the development process, ensuring a user-centric approach that enhances the overall usability and satisfaction of the end product.

Methodology

This research employed an interview-based approach. The interview was conducted in person with Dr. Alice Boone, the end user, who holds a Ph.D. in printing history. The interviewers were undergraduate engineering students, actively engaged in the project, and responsible for its execution. The session took place on January 12 at 9:30 am and extended for approximately one hour. The questions posed during the interview were thoughtfully crafted by the engineering students to comprehensively understand Dr. Boone's requirements, preferences, and contextual considerations relevant to the envisioned hand-printing press. This method allowed for a direct and personalized interaction, fostering a deeper understanding of the user's expectations and contributing valuable insights to guide the subsequent design and development phases of the project.

Results

Dr. Boone wants our team to create a printing press that maximizes error for experimental artists, including children. The press should be adjustable, so it essentially should be able to create different things. The press should create at least one good print, of the size 4 by 6 inches, in which the definition of “good” is up to the interpretation of our team. This interpretation however should be supported by a written justification. Each team can attempt to make their own ink, but it is not required as of this point. The Dr. Boone wants our team to create a printing press that maximizes error for experimental artists, including children. The press should be adjustable, so it essentially should be able to create different things. The press should create at least one good print, of the size 4 by 6 inches, in which the definition of “good” is up to the interpretation

of our team. This interpretation however should be supported by a written justification. Each team can attempt to make their own ink, but it is not required as of this point. The press will reside on Dr. Boone's shelf and should therefore not be too large, but the mechanism is up to each team. Dr. Boone has claimed that we can use her ink and printer paper if needed during the process of creating and trying our press. She has suggested that paper towels help in creating texture and that playdough reproduces images very well. We are able to use kitchen tools, but we can use tools beyond that if necessary.

Discussion

After hearing the results, this project seems to be up to the interpretation of each team to a great extent. We have concluded that our press should not print the same print each time as well. One of the biggest limitations we face is that interpretation is subjective. Our lack of parameters and boundaries for this project may make the team get lost in the goal of what type of press/product we're supposed to be creating. We think it is important to differentiate the perspective of judging art subjectively and objectively, as there are qualities of art we can use to assess if our product is "good enough" rather than relying solely on opinion.

Appendix B

User Observation Summary

Introduction

Team 2, consisting of Mila Hrustic, Crystal Cooper, Andy Lu, And Luke Meyer, met with Dr. Boone Friday morning for an end-user interview. Dr. Boone is the project partner in charge of the printing press parameters. Before experimentation began, she was able to provide us with details on past experiences she has had with printing. We were then able to test different techniques for printing through the usage of different mediums, tools, and pressure application maneuvers. After the creation of approximately 10 prints, we discussed which parts of each print were best and why.

Methodology

Dr. Boone began by giving our team some insight as to how she has printed in the past and what she has prepared today for us to work with. After this brief discussion, Dr. Boone and everyone in the team innovated ways to adapt our prints. During this entire process, Crystal took notes on what Dr. Boone was saying, each experimental process, and the results of each.

Many different adaptations were used when experimenting with how to create different prints. The process began by Luke and Andy carving different designs in foam using an exacto blade. After their designs were made, black ink was placed onto the foam using a brayer before the foam was then pressed onto paper to create a print. Alterations were then made in the usage of different mediums, such as foam core and cardboard, to print on or to print with. The ways to press the ink onto the print’s surface was also manipulated by using even pressure, hands, a burin, and a brayer. Some of the unique objects used to print included metal shavings, twigs, hot glue, honeycomb cardboard, and bubble wrap.

Results

Dr. Boone gave us some background information on printing in our general discussion before experimentation about collagraphs and what has or hasn't worked before. Collagraphs are collages that you print with. She described them as “appealing but also frustrating because you don't know what you're going to get beforehand.” As for what hasn't worked, she claimed potato chips were too curvy and couldn't create the shattered glass pattern she wanted. Two things she mentioned that could help in the printing process, however, are ragged edges of foam core for texture and shellac or spray sealant to make ink stick to surfaces better. Moreover, thinning out ink could be useful since ink leaves prints on its own due to surface tension.

As for the results of the experimenting processes, it seems that using a brayer or burin to apply pressure worked best in even layers of ink on prints. Bubble wrap, hot glue, honeycomb cardboard, and the foam functioned well in reproducing prints, but the twigs, foam core, and metal shavings didn't. As for the order of operations, we found it better to press the foam into the paper rather than vice versa because the paper ended up ripping when peeling it off of the foam. Although mistakes are encouraged, this one was not aesthetic.

When assessing which parts of each print we liked best, we decided that prints with cool textures or unique splotches of ink were most appealing. Additionally, abstract prints that could be interpreted in looking like certain designs were also appealing as they can create interest and conversation.

Discussion

From our experimentation, it was observed that flat surfaces print better, pressing in ink with your hands is the least likely to produce a quality print, different materials require different pressure, and layering on top of less inky prints may be most beneficial for collagraphs.

This meeting was evidence that trial and error will be a huge part of our mockup process. There are a multitude of methods as to how to print, but these experiments gave us a better understanding of what works well and what doesn't. The observation of how Dr. Boone analyzes prints also enables us to view how we should be interpreting the prints we create since we'll have to decide which ones are high quality.

Appendix C

Mockup Testing Summary

Introduction

On February 7th during DTC class, teammates Crystal Cooper, Mila Hrustic, and Andy Lu acted as users to test the mockup they had created. By going to the painting area in the shop within Ford, Team 2 tested the mockups' ability to print using red paint and paper. After this process, the members of this group were able to analyze which parts of the mockups were good and what and how it can be improved.

Methodology

We tested this mockup by brushing paint on the white cardboard and printing onto a paper lined up on the brown cardboard circle on the baseplate. We would then press the 2 faces together. Some questions we asked ourselves were: How can we change the designs on the press? How can we securely fasten the white cardboard? How can we cover a larger area with one press?

Results

Lever stamp allowed for a lot of tests with different types of materials as the changing was simple. The materials were attached onto the stamp using pins, which showed to be a risk as it can cause a smaller injuries. The lever itself was easily movable, however, applying pressure sometimes proved to be more difficult as it had to be done by fingers.

Analysis

The lever stamp had several good qualities. We were able to use different materials and attach them to the stamp portion of the mockup to induce different textures on the paper. You could easily see the patterns and the mockup is pretty simple to operate. However, making complex images that could be reproduced would require further testing.

Appendix D

Executive Summary

Dr. Alice Boone presented our DTC class with the task to create a printing press for inexperienced artists. This press, however, was to be adaptable in the prints it made as it needed to “create discoveries.” As Dr. Boone is most intrigued by art’s mistakes rather than its perfection, she wanted our class to create a press that reflected this concept.

We held a team meeting in the Ford Motor Company Engineering Design Center with Dr. Boone where we observed which materials printed best. The most noteworthy observations consisted of finding that different textures of cardboard, bubblewrap, and rubber bands created abstract designs that were still sharp. We created two mockups based on these observations, tested them out ourselves, and eliminated the rubber band design based on its lack of adjustability and user difficulty. We continued adapting our surviving mockup to be more unique, and eventually created our final prototype after the adjustments were made.

The Printato includes a potato masher with a velcro attachment on its face.The stick of the masher is able to be parallel or perpendicular to the ground depending on the user’s preference. Multiple applications with different textures were made to be attached to the masher and consequently create prints once paint is added. The potato masher is attached to a hinge on a lazy susan so that it can move up and down and side to side. All of this is attached to two wooden bases with a drawer slide rail in between them to allow the masher to move forward and backward. Once the paper is set onto the biggest wooden base, The Printato is ready for function.

- **Adjustability:** The press is easily adjustable as different velcro attachments can be secured onto the potato masher to produce various designs. The differences in amount of force applied when pressing down the masher and numerous possibilities for masher orientation creates variability. Additionally, individuals using this press can attach any other materials to empty velcro attachments if they are looking to create even more designs. They can also change the shape of the attachment by cutting the velcro piece into a shape other than circular, like rectangular.
- **User-Friendly Experience:**
    - **Easy Use:** The simplicity of this design enables the user to create a print with this device without the need for instructions.
    - **Easy to Clean:** A clear plastic adhesive sheet was placed onto the base of the board in which the paper is set on. If paint from the potato masher gets onto the base rather than the paper, a wipe can easily clean this off.
    - **Safe Procedure:** All of the screws used to create this press are secured without any sharp edges poking out. The lack of danger from using this device makes it kid-friendly and suitable for the target audience of inexperienced artists.

It is important to note that this press is geared more towards abstract designs than realistic art pieces.

Appendix E

Instructions for Use

1. Before using the printing press, the user should choose which texture they want to print with (Figure 8). They should attach it on the potato masher with the velcro back (Figure 9).

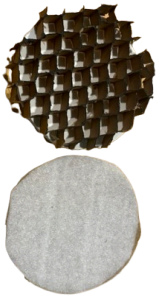


Figure E1. Texture attachments



Figure E2. Potato Masher

2. Once the texture is attached, the user will put the printing paper onto the baseboard of the press (Figure 10).
3. Adjust the drawer slides and lazy susan joint in order to obtain the desired orientation of the press (Figure 11).
4. Put the paint on the texture attached to the potato masher, and press down on the printing paper (Figure 12).

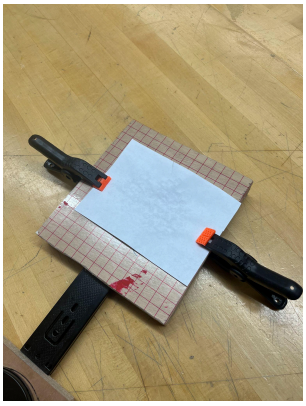


Figure E3. Paper on Baseboard

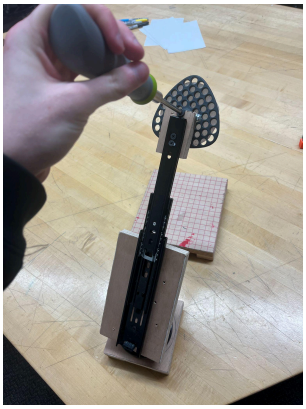


Figure E4. Adjusting Press

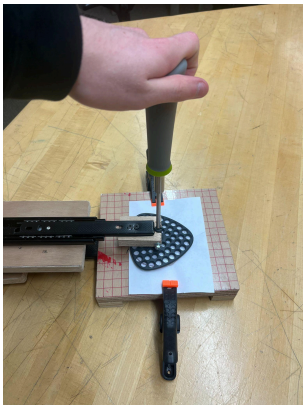


Figure E5. Pressing Down

5. Take out the print, and shorten the drawer slides so it's in its most compact form (Figure 13)



Figure E6. Smallest Configuration

Washing:

1. An adhesive plastic sheet is attached to the wood, so any paint that gets on the wood can come out.
2. Get a towel with some water.
3. Wipe down the baseboard.

Appendix F

Instructions for Construction

Introduction

These instructions will lead you through the process of constructing the potato masher hand press safely and efficiently. The instructions will be divided into the following sections:

- Bill of Materials
- Theory of Operation
- Directions
    - Constructing the baseplate
    - Constructing the printing arm
    - Attaching the lazy susan to the baseplate
    - Attaching the printing arm to the lazy susan
    - Creating print templates/designs
- Troubleshooting

Bill of Materials

Potato Masher
Lazy Susan
2 Drawer Slide Rails
1 inch thick wood sheets
Foam core
Metal door hinge
12 screws
Fluffy dots
Bubblewrap
Cardboard
Honeycomb designed cardboard
Velcro
Acrylic Plastic
Hot glue gun with glue sticks
Drill with appropriate drill bits

Theory of operation

The basic mechanism of our hand press is a printing arm attached to a drawer rail and a lazy susan which allows our arm to have free range of motion over the baseplate. The printing arm is a potato masher screwed onto a drawer rail which is screwed into a wooden plank. The

printing arm is attached to the lazy susan with a door hinge. The lazy susan is glued and screwed onto a wooden plank which is attached to the baseplate by a screwed-on drawer rail. The printing arm can freely rotate and can extend to over 12 inches which allows for versatility.

Instructions:

For reference, the press should like this:



Constructing the baseplate:

1) Cut some 1 inch thick wood to 8 inches in length and 7.5 inches in width.
2) Cut some 1 inch thick wood to 1.5 inches in length and width
3) Use 2 screws to attach the rail in the middle of the 8” x 7.5” wooden plate. Make sure the rail is extended fully when you screw in the screws.



4) Cut out a piece of acrylic plastic to 8 inches in length and 7.5 inches in width.
5) Lay the acrylic on the baseplate (on the opposite side of the rail)

Constructing the printing arm

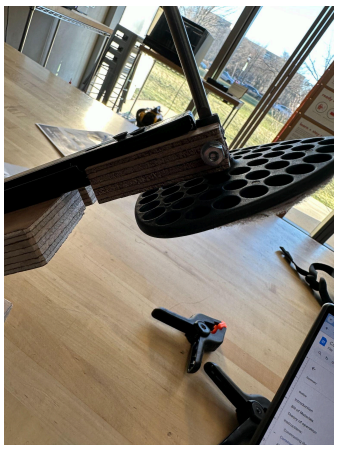
1) Cut some 1 inch thick wood to 8in long and 3.5in wide
2) Use 2 screws to attach a rail at the bottom of the wood. The top of the rail should barely extend over the top of the printing arm.



3) Cut out a piece of foam core as wide as the groove in the rail
4) Cut out a second piece of foam core with near-identical dimensions
5) Cut some 1 inch thick wood to be 3 inches by 1.5 inches. This will be the mounting block
6) Use a dremel tool to make a notch for the potato masher. Make sure the masher fits snugly in the groove.
7) Use a drill to drill a hole in the middle of the potato masher
8) Drill a hole in the mounting block that goes through the groove. Make sure this hole is straight.
9) Attach the mounting block by placing it on top of the foam core blocks and placing the foam core blocks on the inside of the rail. Use 2 screws to screw the mounting block onto the rail. Make sure the open part of the groove is pointing upwards.
10) Insert an appropriately sized screw through the mounting block with the potato masher in the groove. Make sure the potato masher's handle can still move once mounted. Make sure the potato masher lies flat on the baseplate when pressure is applied.

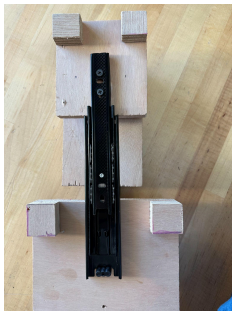
    If mounted properly, the potato masher should be mounted onto the arm like this:





Attaching the lazy susan to the baseplate

1) Cut some 1 inch thick wood into a 5 inch by 4.5 inch rectangle
2) Place the lazy susan in the middle of the wood
3) Use hot glue to glue the lazy susan down in the middle of the wood plate. Do not glue over the hole in the middle of the lazy susan.
4) Use 2 screws and screw the wooden plate to the rail attached to the baseplate. Make sure that the rail is centered on the wood plate with the lazy susan.

5) Cut some 1 inch thick wood into 2 pieces with 1.5 inches in length and width (see image above)

Attaching the printing arm to the lazy susan

1) Use an appropriately sized door hinge and place it on the printing arm so that all 3 screw holes are over the wooden printing arm. Make sure the hinge is on the opposite side of the rail
2) Use 3 screws to screw the hinge to the printing arm
3) Line up the middle hole of the door hinge with the hole in the lazy susan
4) Using a hot glue gun, glue the hinge to the lazy susan while keeping the holes aligned
5) Using one screw, screw the hinge to the wooden plate through the lazy susan.

If mounted properly, it should look like this:



Creating print templates/designs

1) Cut a strip of adhesive velcro in a circle of any size you desire
2) Attach the velcro circle onto the potato masher
3) Cut another strip of velcro in a similar sized circle
4) Using hot glue, attach any sort of pattern, texture or design on the opposite side of the velcro.

Troubleshooting

If a screw is not screwing in, try using an electric screwdriver or a drill with a screw tip.

If the potato masher is not lying flat on the baseplate when pressed down, unscrew the potato masher, place it flat on the baseplate, and then screw it back in.

If the potato masher cannot cover the whole baseplate, unscrew the base plate rail and move it back.

Appendix G

Project Definition

Project Name: Making a printing press out of a common object.
Client: Dr. Alice Boone
Team Members: Luke Meyer, Crystal Cooper, Mila Hrustic, Andy Lu
Date: 3/11/24

Mission Statement: We aim to create a hand printing press for inexperienced artists, including children, that can be adjusted to create “mistakes” that lead to interesting artistic discoveries. While usually bad, these mistakes in the prints can allow users to make unique prints that they otherwise would not have thought to produce.

Deliverables:

- A functioning hand printing press prototype that fits specifications
- A model print
- A report on our prototype
- A formal presentation and poster on our process

Constraints:

- The printing press cannot be made out of an already existing printing press or in a way that is easily discoverable online
- $150 budget
- Must be ready for presentation by March 9th

Users/Stakeholders:

- Dr. Alice Boone = project partner.
- Artists who would like to use a creative way to print.

- Children who are learning about the intersection between engineering and art.

User Profile

Dr. Alice Boone is a manager of the Art + Engineering Initiatives program at Northwestern University. She works to improve the collaboration between engineering and the arts. She mentioned that apart from her, the users of this design would be artists ranging in age from childhood to adulthood. According to Dr. Boone, current hand printing presses are too efficient at producing the desired print. They don't allow any deviation from the person behind the design. This makes it hard to discover new compositions. Dr. Boone is looking for a printing press that is made in a creative way, produces unique prints each time, and is simple enough that any child user would have no difficulty figuring it out.

Requirements	**Specification**
The press should be able to put an image onto a paper	● The paper should be 4±1 by 6±1 inches ● The print must be easily adjustable to print on all parts of the paper
The printing press should be able to fit on a shelf	● Must not exceed 20 lb ● Average medium bookshelf dimensions are 10"-12" in depth, 9"-12" in height for each section, and width of 31"-32"
The print should be easy to clean	● All surfaces exposed to ink of the press must not have barriers to make reaching them hard. ● The ink/color must be washed off with regular water + soap
The press must encourage randomness and variability in its product/the print	● The texture of the image must be changeable or different from a regular print ● The press must produce a different image for the same process

Table G1. Requirements

Appendix H

Secondary Research Summary

At the beginning of the project, we did background research on the subject matter mentioned by our project partner in the description of the project. The description involved the designing and creation of a hand-printing press. The project partner specifically requested that we use a common household item as the base for the hand-printing press. In order to approach an interview with more precise questions, we did some research into the subject matter of the printing press beforehand. We looked into what a hand-press is, vocabulary that would be useful to us, materials that we can print with, and who Dr. Boone is.

What is a hand-press?

A hand-press is simply a printing press that is operated by hand. A printing press is a machine that transfers images onto a variety of materials.



Figure H1. Hand Printing Press
Source: https://www.dickblick.com/products/richeson-hand-printing-press/

Printing Vocabulary/Knowledge

Two terms that consistently come up when researching printing presses are resolution and DPI. Resolution refers to the detail in an image. It can be measured in DPI, otherwise known as dots per inch [12]. The term dots can be thought of as similar to pixels on a computer screen. A higher DPI indicates a higher resolution image. Additionally, when formatting images, make sure to write it as width x length.

What can we print with?

Throughout history, a variety of different methods to print have come into existence. As such the number of materials that can be used to print have increased along with those methods. We researched a few such materials.

Proteins

Even something as small as proteins can be used as a material to print patterns. Using a technique called soft-lithography, proteins or cells can be printed onto a soft surface such as a gel [5]. This is done by mixing ink and the proteins and then stamping that ink solution onto the soft surface.

3D printing lab on a chip devices allows for less sample material and shorter reaction times

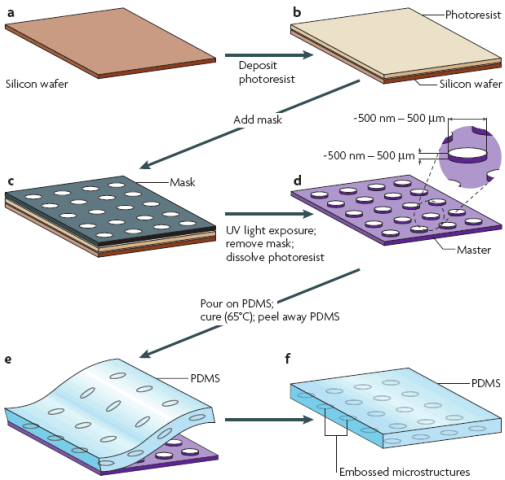


Figure H2. Soft Lithography Process
Source:
https://www.elveflow.com/microfluidic-reviews/soft-lithography-microfabrication/introduction-about-soft-lithography-and-polymer-molding-for-microfluidic/

Food

There are also a number of ways to print on or with food as well. Using edible ink, it is possible to create images on chocolate or other food surfaces [7]. We discovered that you can also print images by using vegetables and fruit as stamps. There is even the possibility of using food like chocolate as a material in 3D printing since that form of printing is so versatile in its materials.

Who is Dr. Boone?

As she was the person that this project is going to be for, it was important to understand Dr. Boone so that we can understand her requests. Dr. Boone is directly involved in the Artists-at-Large Program and Art + Engineering Initiatives at Northwestern University [1]. As such, Dr. Boone works in the intersection of engineering and art, which is why the request for a hand-printing press made in unconventional ways is expected.

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